Atharva Aalok

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EDUCATION

MIT — On Leave of Absence
- Doctor of Philosophy, Mechanical Engineering — GPA: -/-
- **Advisor:** Prof. Faez Ahmed
- **Dual Admit:** Computational Science and Engineering (CSE) PhD Program
- **Award:** Department of Mechanical Engineering Graduate Fellowship

Stanford University — Graduation: June 2026
- Master of Science, Aeronautical and Astronautical Engineering — GPA: 4.07/4.0
- **Advisor:** Prof. Juan J. Alonso

Indian Institute of Technology (IIT), Madras — Graduation: May 2023
- Bachelor of Technology, Aerospace Engineering — CGPA: 9.38/10
- **Advisor:** Prof. R.I. Sujith
- **Department Rank:** *1/58*
- **Awards:** Hindustan Aeronautics Limited (HAL) Prize, Institute Silver Medal

WORK EXPERIENCE

Godela (YC X25) — April 2026 - Present
Founding Engineer — *godela.ai*
- Designed and built Godela's agentic geometry and CAD pipelines from scratch, end to end.
- Achieved state-of-the-art performance on several benchmarks, ranking #1 on the CADGenBench leaderboard on Hugging Face and on MIT's CADBench.
- Work closely with Godela's design partners, including Nvidia, Dell, Dallara and Hark.
- Founded the testing team and worked on both the frontend and backend of Godela's agentic platform.

PUBLICATIONS

Atharva Aalok, Juan Alonso, "NIGnets: Neural Injective Geometry Networks for Representing non-self-intersecting Geometry". Link

Jiayi Zhou*, **Atharva Aalok***, Valentin Duruisseaux, Xinyi Li, Juan Alonso, Anima Anandkumar. "Robust Shape Optimization with Neural Shape Representations and Neural Operators".

Atharva Aalok, Juan Alonso. "Robust Aerodynamic Shape Optimization via Deep Reinforcement Learning with Non-Self-Intersecting Geometry Networks". Link

Atharva Aalok, Induja Pavithran, R.I. Sujith. "Breaking points of Early Warning Signals: Robustness in Rate-delayed tipping regime". **Accepted in** Conference on Nonlinear Systems and Dynamics.

RESEARCH EXPERIENCE

Robust ML based Shape Optimization — Nov 2024 - March 2026
Prof. Juan J. Alonso (Stanford), Collaborator: Prof. Anima Anandkumar (Caltech)
- Robust shape optimization using Neural Operators (FNOGNO, GINO, BNO) with geometry represented using NIGnets and NeuralODEs to prevent self-intersection and optimization over in-distribution shapes.
- Generated Euler and RANS datasets with ∼20,000 simulations each using SU2 for 1500 airfoils at 16 different angles of attack. Trained Neural Operators to 2% accuracy on pressure field and skin-friction predictions.
- Trained a NeuralODE based generative geometry model with hard guarantees on non-self-intersection. Achieved extremely accurate shape reconstruction loss of 1e-8 L2 on the entire shape dataset.

NIGnets: Neural Injective Geometry Networks Sept 2024 - December 2025

Prof. Juan J. Alonso, Aerospace Design Lab, Stanford University

- Developed a new neural architecture that gives a hard guarantee on representing only non-self-intersecting geometry.
- Utilized monotonic networks for expressivity, replacing activation functions, while maintaining injectivity.
- Developed the concept of Pre-Auxilliary networks to allow NIGnets to work as generative models, parameterizing non-self-intersecting shapes using latent vectors.
- Paper, Github, Website, Presentation

Shape Optimization using Deep Reinforcement Learning March 2025 - October 2025

Prof. Juan J. Alonso (Stanford), Collaborator: Prof. Elie Hachem (CEMEF Mines Paris - PSL)

- Used NIGnets and NICE as our geometry representation to reduce the design space that agents search through by avoiding unphysical (self-intersecting) shapes.
- Trained PPO and AAC based agents to maximize L/D ratio using XFoil as flow solver. Used Hindsight Experience Replay to improve sample efficiency as well as design for a specified L/D ratio.
- Achieved L/D ratios upto 61.64 and trained policies to design for specified L/D ratios. Paper, Github

DATASETS

RANS - Subsonic June 2025 - March 2026

Prof. Juan J. Alonso, Stanford University

- ∼250,000 RANS simulations of 1500 UIUC airfoils at 16 different angles of attack from Mach 0.01-0.5.
- CFD using SU2 with meshing done in Gmsh. Computations done on the Stanford HPC cluster.
- Expansion and curation in-progress. Dataset on Google Cloud.

TEACHING EXPERIENCE

AA141 - Atmospheric Flight (Aircraft Conceptual Design) April - June 2025

Prof. Juan J. Alonso, Stanford University

- Held weekly office hours to clarify concepts and assist students with coursework.
- Graded assignments and provided feedback to support student learning.

UNDERGRADUATE RESEARCH EXPERIENCE

Design of Jet Based UAVs with Thrust Vectoring May 2022 - Jan 2023

Prof. S. Sundaram, Artificial Intelligence and Robotics Lab, IISc Bangalore

- Designing a high altitude (5000m+) jet based UAV with 50kg payload capacity and STOL capabilities.

  Phase 2
  - Worked towards building a prototype vehicle as proof of concept of a novel 2-DOF thrust vectoring mechanism.
  - Design: Prototype.

  Phase 1
  - Modelled a 3-DOF thrust stand and fabricated through laser cutting, CNC machining, milling and 3D printing. Chose and calibrated temperature, pressure and force sensors and set up data acquisition. Handled purchase worth over 10 Lakh ($12k).

Robustness of Early Warning Signals June 2021 - November 2022

Prof. R.I. Sujith, Sujith's Lab, IIT Madras

- Rigorously studied the behavior of Early Warning Signals (EWSs) for a 3rd order non-autonomous power system model. Discovered emergence of rate-induced tipping in our system. Studied EWS for predicting rate-delayed and rate-induced tipping.
- Accepted in Conference on Nonlinear Systems and Dynamics.

Analytical Investigations in Rate-Induced Tipping – Bachelor's Thesis August 2022 - June 2023
Prof. R.I. Sujith, Sujith's Lab, IIT Madras

- Derived a theoretical result governing the relative motion between the system state and its fixed points in phase space for a general non-autonomous n^{th} order dynamical system and showed that it can be analyzed from a control system perspective.

OTHER PROJECTS

DeepFusion - Built a deep learning framework from scratch! MLPs, CNNs, RNNs etc. 2024

NIGnets - Library for new neural architecture we built for hard guarantees on non-self-intersecting geometry. 2025

GeoDiff - A differentiable geometry representation library. (Work in progress) 2025

GeoSimilarity - A differentiable geometric loss function library. 2025

Moving Window Statistics - Matlab toolbox to compute moving window statistics fast! Speed-up 50-760x. 2023

Professional Plots - Matlab toolbox for creating publication quality plots quickly. Over 4200 downloads! 2023

Search and Rescue UAV Design - Designed a 10.47kg, 1.95m wingspan UAV design with Payload capacity: 1.3kg, Range: 115km, Endurance: 90 min and Cruise Speed: 20m/s. 2022

TECHNICAL SKILLS

Programming Languages	:	Python, MATLAB, C/C++
CFD	:	SU2, Luminary Cloud, Ansys, OpenFOAM, Gmsh
Machine Learning	:	PyTorch, JAX, Stable Baselines3
Hardware	:	Raspberry Pi, Arduino, Thermocouples, Pitot tubes, Load Cells
3D Modelling	:	SolidWorks, Fusion 360, OnShape

AWARDS AND RECOGNITION

- Department of Mechanical Engineering Graduate Fellowship, MIT. 2026
- HAL Prize (Institute Silver Medal), IIT Madras (best academic performance in Aerospace Engineering). 2023
- Recipient of the prestigious KVPY Fellowship by Dept. of Science and Technology, Govt. of India. 2019
- Awarded the coveted National Talent Search Examination (NTSE) fellowship by NCERT, Govt. of India 2018
- Amongst the top 2% Open-source contributor on MathWorks (MATLAB Community). Present

EXTRA-CURRICULAR ACTIVITIES

- Department Academic Legislator, Elected representative, Department of Aerospace Engineering, IIT Madras 2022-2023.
- Member, Aerospace Engineering Association, IIT Madras 2022-2023.
- Freshman Mentor, SAATHI - IITM Mental Health and Wellness Community, IIT Madras 2021-2022.
- Long distance events, running: half marathons and cycling: Gran Fondo (100km+) rides with participation in marathons in multiple cities.